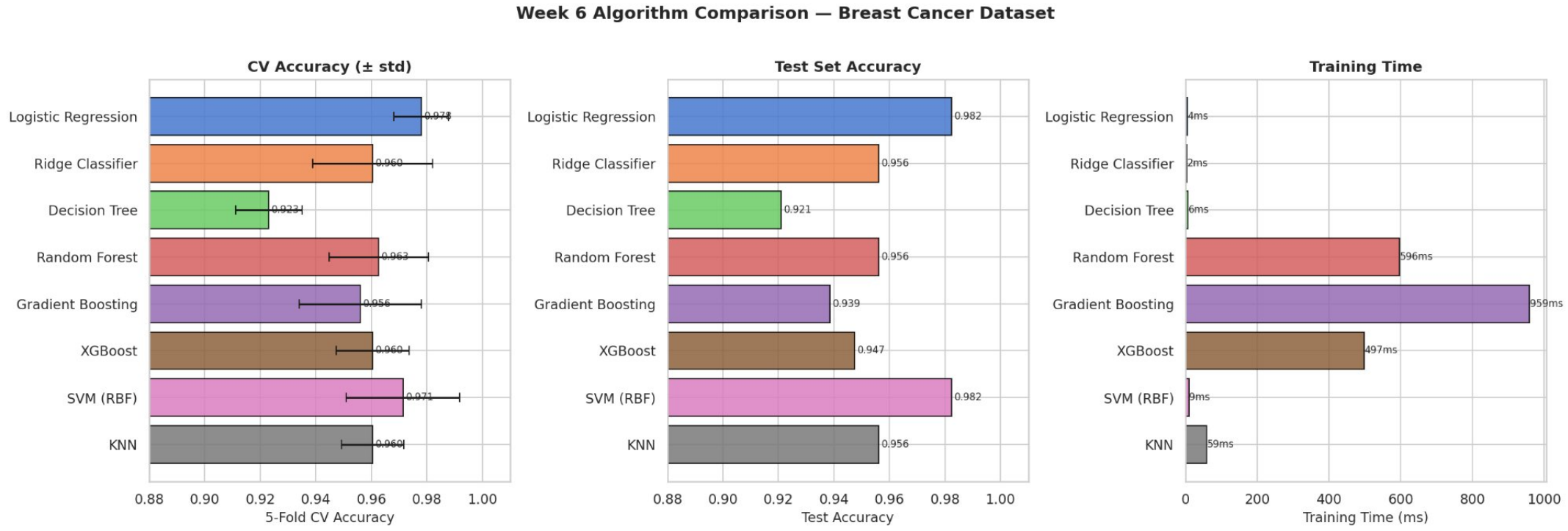
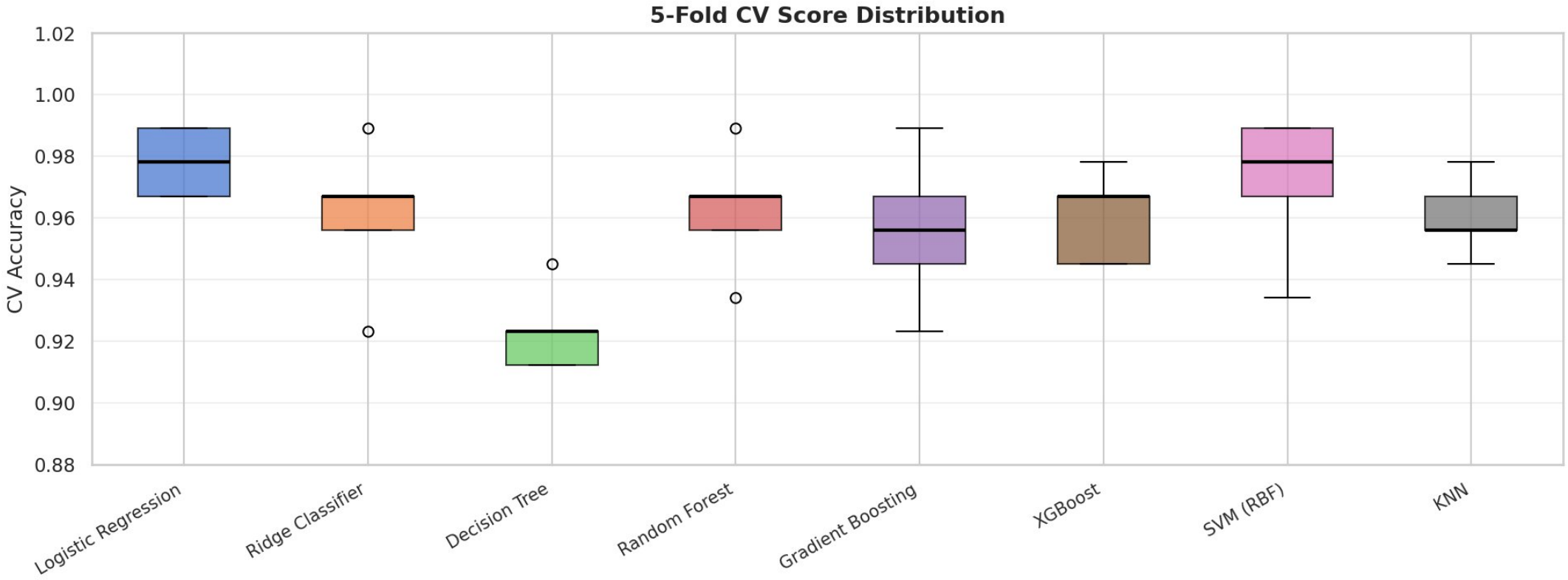
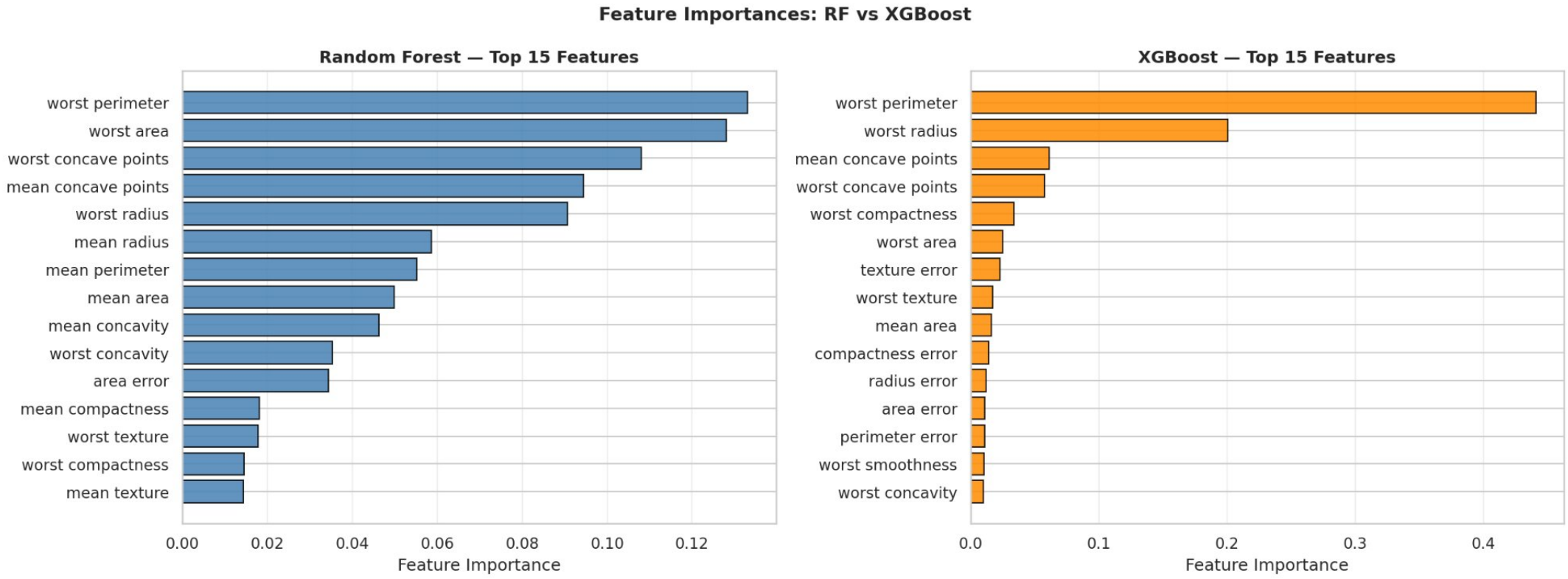
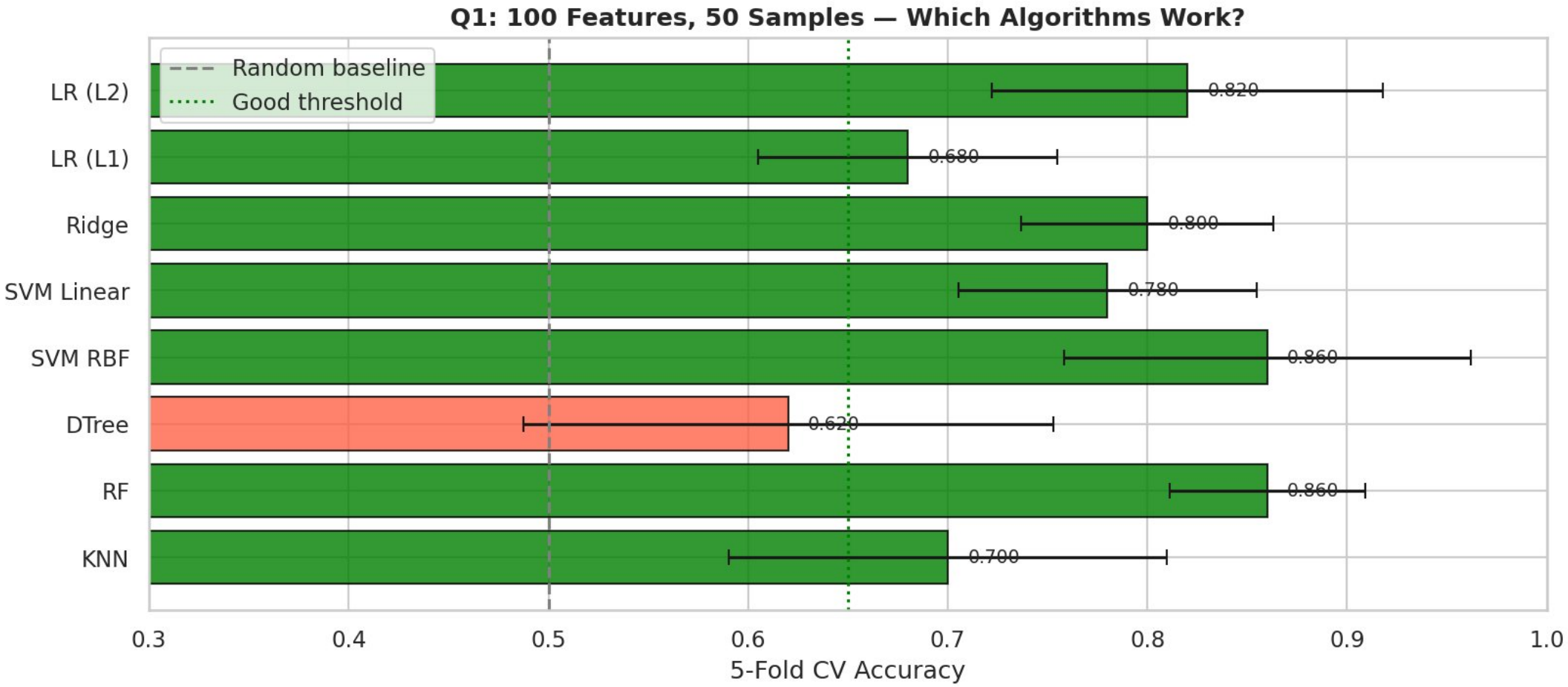

Week 6 Algorithm Comparison — Breast Cancer Dataset
CV Accuracy (± std)
Logistic Regression
Ridge Classifier
Decision Tree
Random Forest
Gradient Boosting
XGBoost
SVM (RBF)
KNN
0.978
0.960
0.923
0.963
0.956
0.960
0.971
0.960
0.88
0.90
0.92
0.94
0.96
0.98
1.00
5-Fold CV Accuracy
Test Set Accuracy
0.982
0.956
0.921
0.956
0.939
0.947
0.982
0.956
Test Accuracy
Training Time
4ms
2ms
6ms
596ms
959ms
497ms
9ms
59ms
0
200
400
600
800
1000
Training Time (ms)

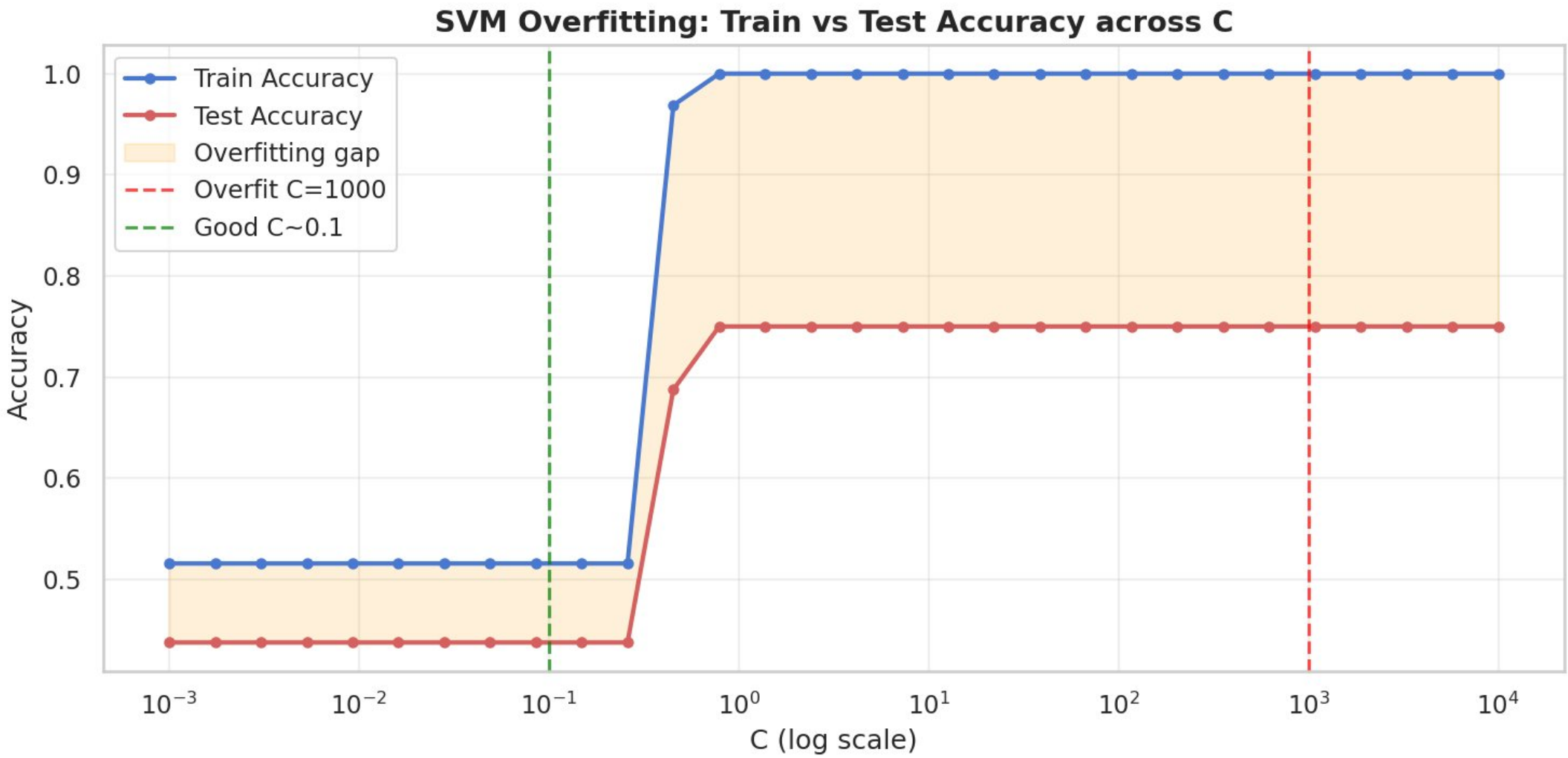
Algorithm Selection Guide — Week 6 ML

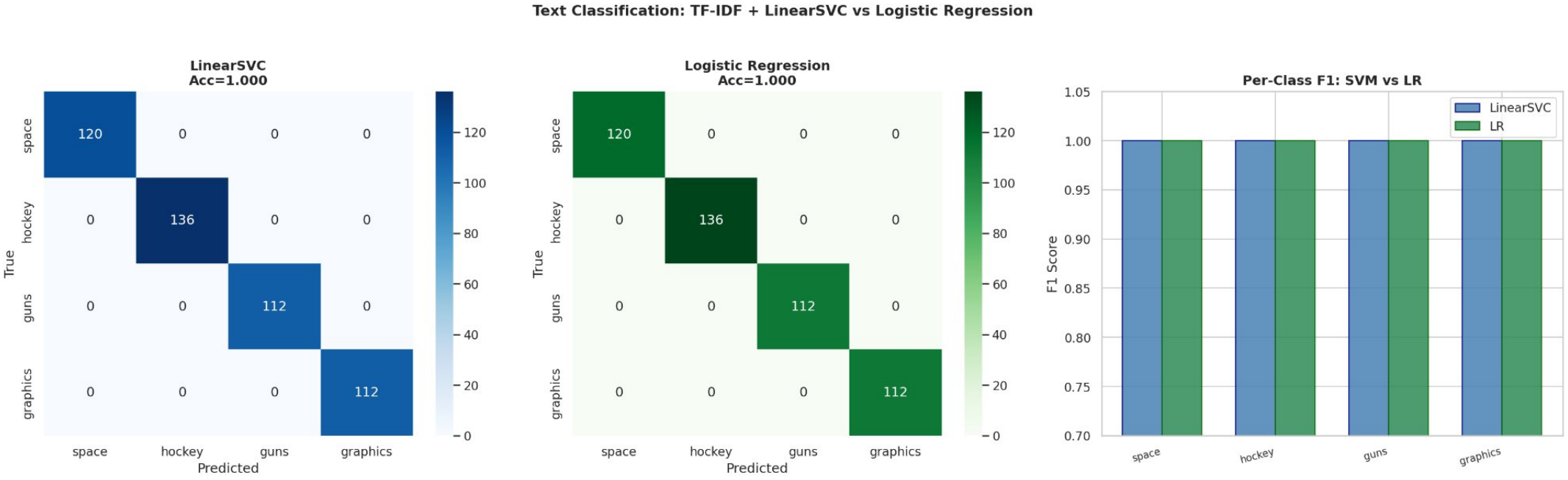
Scenario	Best Choice	Why	Avoid
Text / NLP (TF-IDF)	LinearSVC / LR	Sparse high-dim space	RF, KNN
n < 200, d > 100	LR (L1), SVM(linear)	Regularisation prevents overfit	XGB, KNN
Tabular, max acc	XGBoost	SOTA on tabular data	KNN
Need interpretability	Logistic Regression	Coefficients = feature impact	RF, SVM
Missing values	XGBoost	Native handling	KNN, SVM
Non-linear boundary	RF or SVM (RBF)	Non-parametric / kernel	LR, Ridge
Need probabilities	LR or RF	Calibrated outputs	SVM, Ridge
Large n > 50K	LR, LinearSVC	Scales linearly	SVM(RBF), KNN
Fast robust baseline	Random Forest	Low variance, parallel	DT alone
No feature scaling	DT, RF, XGBoost	Tree splits are scale-free	KNN, SVM, LR

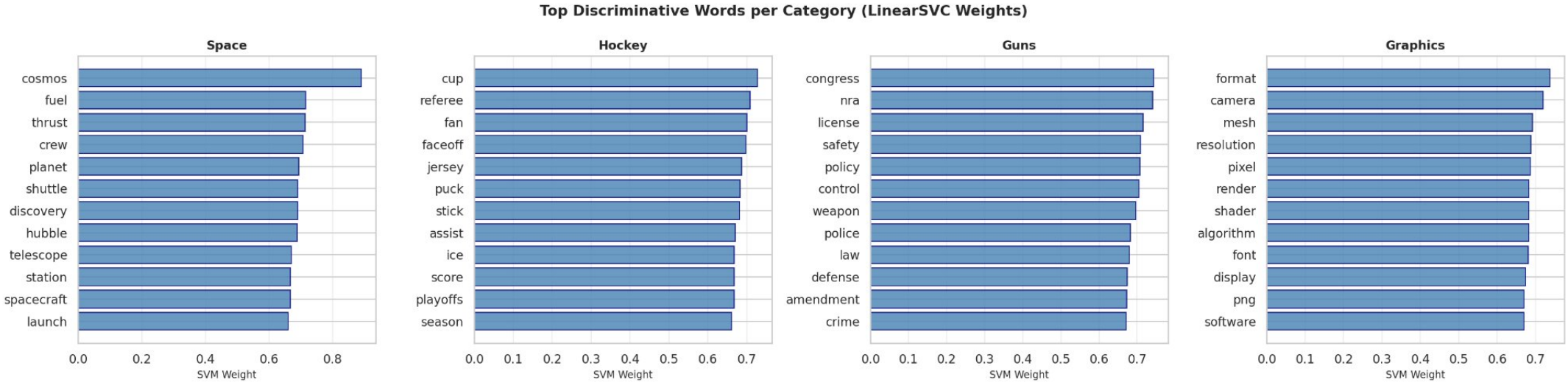

5-Fold CV Score Distribution
CV Accuracy
1.02
1.00
0.98
0.96
0.94
0.92
0.90
0.88
Logistic Regression
Ridge Classifier
Decision Tree
Random Forest
Gradient Boosting
XGBoost
SVM (RBF)
KNN


Feature Importances: RF vs XGBoost
Random Forest — Top 15 Features
worst perimeter
worst area
worst concave points
mean concave points
worst radius
mean radius
mean perimeter
mean area
mean concavity
worst concavity
area error
mean compactness
worst texture
worst compactness
mean texture
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Feature Importance
XGBoost — Top 15 Features
worst perimeter
worst radius
mean concave points
worst concave points
worst compactness
worst area
texture error
worst texture
mean area
compactness error
radius error
area error
perimeter error
worst smoothness
worst concavity
0.0
0.1
0.2
0.3
0.4
Feature Importance


Q1: 100 Features, 50 Samples — Which Algorithms Work?
Random baseline
Good threshold
LR (L2)
LR (L1)
Ridge
SVM Linear
SVM RBF
DTree
RF
KNN
0.820
0.680
0.800
0.780
0.860
0.620
0.860
0.700
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
5-Fold CV Accuracy


SVM Overfitting: Train vs Test Accuracy across C
Train Accuracy
Test Accuracy
Overfitting gap
Overfit C=1000
Good C~0.1
Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
C (log scale)


Text Classification: TF-IDF + LinearSVC vs Logistic Regression
LinearSVC
Acc=1.000
True
Predicted
space
hockey
guns
graphics
120
0
0
0
0
136
0
0
0
0
112
0
0
0
0
112
Logistic Regression
Acc=1.000
True
Predicted
space
hockey
guns
graphics
120
0
0
0
0
136
0
0
0
0
112
0
0
0
0
112
Per-Class F1: SVM vs LR
F1 Score
LinearSVC
LR
1.05
1.00
0.95
0.90
0.85
0.80
0.75
0.70
space
hockey
guns
graphics


Top Discriminative Words per Category (LinearSVC Weights)
Space
cosmos
fuel
thrust
crew
planet
shuttle
discovery
hubble
telescope
station
spacecraft
launch
0.0 0.2 0.4 0.6 0.8
SVM Weight
Hockey
cup
referee
fan
faceoff
jersey
puck
stick
assist
ice
score
playoffs
season
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
SVM Weight
Guns
congress
nra
license
safety
policy
control
weapon
police
law
defense
amendment
crime
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
SVM Weight
Graphics
format
camera
mesh
resolution
pixel
render
shader
algorithm
font
display
png
software
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
SVM Weight